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Question 1. With a Fine-Truth Comb

(a)

P	Q	$P \leftrightarrow Q$	$\neg P$	$P \wedge Q$	$\neg P \leftrightarrow (P \wedge Q)$
F	F	T	T	F	F
F	T	F	T	F	F
T	F	F	F	F	T
T	T	T	F	T	F

$P \leftrightarrow Q \not\equiv \neg P \leftrightarrow (P \wedge Q)$ because the results of the expressions differ in rows 1, 2, and 4.

(b)

P	Q	R	$P \leftrightarrow Q$	$(P \leftrightarrow Q) \rightarrow R$	$P \vee R$	$Q \rightarrow R$	$(P \vee R) \wedge (Q \rightarrow R)$
F	F	F	T	F	F	T	F
F	F	T	T	T	T	T	T
F	T	F	F	T	F	F	F
F	T	T	F	T	T	T	T
T	F	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	T	F	T	F	T	F	F
T	T	T	T	T	T	T	T

$(P \leftrightarrow Q) \rightarrow R \not\equiv (P \vee R) \wedge (Q \rightarrow R)$ because the results of the expressions differ in row 3.

Question 2. Too Cool For Rule

(a)

$$\begin{aligned}
 (\neg P \vee Q) \wedge (\neg P \wedge Q) &\equiv ((\neg P \vee Q) \wedge \neg P) \wedge Q && \text{Associativity} \\
 &\equiv (\neg P \wedge (\neg P \vee Q)) \wedge Q && \text{Commutativity} \\
 &\equiv \neg P \wedge Q && \text{Absorption}
 \end{aligned}$$

(b)

$$\begin{aligned}
 (P \rightarrow \neg Q) \wedge (\neg P \rightarrow \neg Q) &\equiv (P \rightarrow \neg Q) \wedge (Q \rightarrow P) && \text{Contrapositive} \\
 &\equiv (\neg P \vee \neg Q) \wedge (Q \rightarrow P) && \text{Law of Implication} \\
 &\equiv (\neg P \vee \neg Q) \wedge (\neg Q \vee P) && \text{Law of Implication} \\
 &\equiv (\neg Q \vee \neg P) \wedge (\neg Q \vee P) && \text{Commutativity} \\
 &\equiv \neg Q \vee (\neg P \wedge P) && \text{Distributivity} \\
 &\equiv \neg Q \vee (P \wedge \neg P) && \text{Commutativity} \\
 &\equiv \neg Q \vee F && \text{Law of Non-Contradiction} \\
 &\equiv \neg Q && \text{Domination}
 \end{aligned}$$

(c)

$$\begin{aligned}
 (P \vee Q \rightarrow R) \rightarrow Q &\equiv (\neg(P \vee Q) \vee R) \rightarrow Q && \text{Law of Implication} \\
 &\equiv \neg(\neg(P \vee Q) \vee R) \vee Q && \text{Law of Implication} \\
 &\equiv (\neg\neg(P \vee Q) \wedge \neg R) \vee Q && \text{De Morgan's Law} \\
 &\equiv ((P \vee Q) \wedge \neg R) \vee Q && \text{Double Negation} \\
 &\equiv Q \vee ((P \vee Q) \wedge \neg R) && \text{Commutativity} \\
 &\equiv (Q \vee P \vee Q) \wedge (Q \vee \neg R) && \text{Distributivity} \\
 &\equiv (P \vee Q \vee Q) \wedge (Q \vee \neg R) && \text{Commutativity} \\
 &\equiv (P \vee Q) \wedge (Q \vee \neg R) && \text{Idempotency} \\
 &\equiv (P \vee Q) \wedge (\neg R \vee Q) && \text{Commutativity} \\
 &\equiv (\neg\neg P \vee Q) \wedge (\neg R \vee Q) && \text{Double Negation} \\
 &\equiv (\neg P \rightarrow Q) \wedge (\neg R \vee Q) && \text{Law of Implication} \\
 &\equiv (\neg P \rightarrow Q) \wedge (R \rightarrow Q) && \text{Law of Implication}
 \end{aligned}$$

Question 3. Truth or Feb?**(a)**

- i. Every month with an even number of days has exactly 30 days.

This is false because February can have 28 days.

- ii. Every month with an even number of days, excluding February, has exactly 30 days.

This is true because it handles the case of February.

- iii. If every month has an even number of days and isn't February, then every month also has exactly thirty days.

This is true because the premise is false - February exists.

(b)**(c)**

- i. $\neg \exists x (\text{HasThirtyDays}(x) \wedge \text{IsFebruary}(x))$
ii. $\exists x (\neg \text{HasEvenDays}(x) \wedge \neg \text{IsFebruary}(x))$
iii. $\text{HasEvenDays}(x) \wedge \neg \text{HasThirtyDays}(x) \rightarrow \text{IsFebruary}(x)$